ECON 770 Recitation 9

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Definition & Notations

- Random vector $X = (X_1, X_2, \ldots, X_k)' \in \mathbb{R}^{k\times 1}$, $k \in \mathbb{N}$
- Inner product $X'X = \sum_{i=1}^{k} X_i^2$
- Euclidean norm $||X||_2 = (X'X)^{1/2} = \sqrt{\sum_{i=1}^{k} X_i^2}$

  L^1-norm $||X||_1 = \sum_{i=1}^{k} |X_i|$

  L^∞-norm $||X||_\infty = \max\{X_1, \ldots, X_k\}$
- PDF: let random vector $X \sim f(x)$ s.t.

$$\int f(x)dx = \underbrace{\int \cdots \int}_{k} f(x_1, \ldots, x_k)dx_1 \cdots dx_k = 1$$

- Marginal PDF:

$$f_{X_1}(x_1) = \underbrace{\int \cdots \int}_{k-1} f(x_1, \ldots, x_k)dx_2 \cdots dx_k$$

Definition & Notations (Contd)

- $\mathbb{E}[X_1] = \int x_1 f_{X_1}(x_1) dx_1$ (or let $u = x_1$, just notation)
- $\mathbb{E}[X] = (\mathbb{E}[X_1], \ldots, \mathbb{E}[X_k])'$ (element-wise)
- CDF: $X \sim F(x)$ then

$$\begin{aligned} F(x) = P(X \leq x) =& P(X_1 \leq x_1, \ldots, X_k \leq x_k) \\ =& \int_{-\infty}^{x_1} \cdots \int_{-\infty}^{x_k} f(u_1, \ldots, u_k) du_1 \cdots du_k \end{aligned}$$

- Elements $X_1, \ldots, X_k$ are independent iff

$$P(X \leq x) = \prod_{i=1}^{k} P(X_i \leq x_i) \quad \text{iff} \quad f(x) = \prod_{i=1}^{k} f(x_i)$$

Definition & Notations (Contd)

- Variance/Covariance Matrix

$$\Sigma = \begin{pmatrix} \mathbb{E}\left[(X_1-\mu_1)^2\right] & \cdots & \mathbb{E}\left[(X_1-\mu_1)(X_k-\mu_k)\right] \\ \vdots & \ddots & \vdots \\ \mathbb{E}\left[(X_k-\mu_k)(X_1-\mu_1)\right] & \cdots & \mathbb{E}\left[(X_k-\mu_k)^2\right] \end{pmatrix}$$

or we can write it as

$$\Sigma = E\left[(X-\mu)(X-\mu)'\right] = \begin{pmatrix} \sigma_1^2 & \cdots & \sigma_{1,k} \\ \vdots & \ddots & \vdots \\ \sigma_{k,1} & \cdots & \sigma_k^2 \end{pmatrix}$$

- Σ is symmetric and positive semi-definite (psd)

Finite Expectation

Theorem

Given random vector $X = (X_1, X_2, \ldots, X_k)$, we know

$$\mathbb{E}[|X|] < \infty \quad \Leftrightarrow \quad \mathbb{E}[|X_i|] < \infty \; \forall \, i = 1, \ldots, k$$

where $|X| = \sum_{i=1}^{k} |X_i|$.

What is norm?*

A vector space norm on K-vector space V is a function $|| \cdot || : V \times V \longrightarrow \mathbb{R}$ s.t.

- $||v|| \geq 0$ for all $v \in V$ and $||v|| = 0$ if and only if $v = 0$
- $||v + w|| \leq ||v|| + ||w||$ for all v and w in V
- $||cv|| = |c| \cdot ||v||$ for all $c \in \mathbb{R}^{dim(V)}$ and $v \in V$

CDF and PDF

Consider continuous random vector $V = (X, Y, Z)$ with CDF F_V and PDF f_V, then

$$F_V(v) = P(V \leq v) = F_V(x, y, z) = \int_{-\infty}^{z} \int_{-\infty}^{y} \int_{-\infty}^{x} f_V(i, j, k) \mathrm{d}i \mathrm{d}j \mathrm{d}k$$

By Leibniz's rule $\frac{\partial}{\partial x} \int_{-\infty}^{x} f(a) da = f(x) \times 1 = f(x)$,

$$\frac{\partial^3}{\partial x \partial y \partial z} F_V(x, y, z) = \frac{\partial^2}{\partial x \partial y} \left(\frac{\partial}{\partial z} \int_{-\infty}^{z} \int_{-\infty}^{y} \int_{-\infty}^{x} f_V(i, j, k) \mathrm{d}i \mathrm{d}j \mathrm{d}k \right) = f_V(x, y, z)$$

Best Linear Predictor

- Assume we have random vector $(X, Y) \sim f(x, y)$, and we want to find the best[1] linear predictor $P(Y|X) = B^{*\prime}X$, where $B, X \in \mathbb{R}^{k\times 1}$ and $Y \in \mathbb{R}$
- To solve the optimal B_*, by FOC,

$$\frac{\partial}{\partial B}\mathbb{E}\left[(Y - B'X)^2\right] = -2\mathbb{E}\left[(Y - B'X)X\right] = 0$$

  which implies[2] $B^* = (\mathbb{E}\left[XX'\right])^{-1}\mathbb{E}\left[XY\right]$
- When we play with data, i.e. $Y \in \mathbb{R}^{n\times 1}, X \in \mathbb{R}^{n\times k}$, then $P(Y|X) = XB^*$

$$B^* = (X'X)^{-1}X'Y$$

[1] Optimal coefficients B^* s.t. Mean-Squared Error (MSE) $\mathbb{E}\left[(Y - B'X)^2\right]$ minimized

[2] We need invertible (or nonsingular/full-rank/not-perfect-linear-correlated) $\mathbb{E}\left[XX'\right]$

Best MSE Predictor

Theorem

Conditional expectation $g(X) = \mathbb{E}[Y|X]$ is the best MSE predictor, i.e. for any function $h(X)$,

$$\mathbb{E}\left[(Y - g(X))^2\right] \leq \mathbb{E}\left[(Y - h(X))^2\right]$$

Let $\varepsilon = Y - g(X)$, then we know $\mathbb{E}[\varepsilon|X] = \mathbb{E}[Y|X] - \mathbb{E}[g(X)|X] = 0$, which implies (by LIE)

- $\mathbb{E}[\varepsilon] = 0$
- $\mathbb{E}[\varepsilon f(X)] = 0$ for any function $f(X)$

i.e. ε are uncorrelated (linearly orthogonal) with any function of X

Best MSE Predictor (Proof)

Proof

Since $\mathbb{E}\left[(Y-h(X))^2\right] = \mathbb{E}\left[(Y-g(X)+g(X)-h(X))^2\right]$

$$= \underbrace{\mathbb{E}\left[(Y-g(X))^2\right]}_{\geq 0} + \underbrace{\mathbb{E}\left[(g(X)-h(X))^2\right]}_{\geq 0} + 2\mathbb{E}\left[(Y-g(X))(g(X)-h(X))\right],$$

by LIE,

$$\begin{aligned}\mathbb{E}\left[(Y-g(X))(g(X)-h(X))\right] &= \mathbb{E}\left[\mathbb{E}\left[(Y-g(X))(g(X)-h(X)) \mid X\right]\right] \\ &= \mathbb{E}\left[(g(X)-h(X))\mathbb{E}\left[Y-g(X) \mid X\right]\right] \\ &= \mathbb{E}[(g(X)-h(X))(\underbrace{\mathbb{E}\left[Y|X\right]-g(X)}_{=0})]\end{aligned}$$

Therefore,

$$\mathbb{E}\left[(Y-h(X))^2\right] = \mathbb{E}\left[(Y-g(X))^2\right] + \mathbb{E}\left[(g(X)-h(X))^2\right] \geq \mathbb{E}\left[(Y-g(X))^2\right]$$

Best MSE Predictor (Contd)

Theorem

- $\mathbb{E}[Y|X]$ is unique (almost surely)
- If $\mathbb{E}[Y|X] = a + b'X$ and $\mathbb{E}\left[(X - \mu_X)(X - \mu_X)'\right]$ is positive definite[a], then parameters (a, b) are unique (Hint: we can solve them explicitly)

[a] Invertible covariance matrix

Proof: See lecture notes or the last question in the midterm.

Multivariate Normal Distribution

- Bivariate normal distribution $(X_1, X_2) \sim N(\mu_1, \mu_2, \sigma_1^2, \sigma_2^2, \rho)$ or

$$N(\boldsymbol{\mu}, \boldsymbol{\Sigma}) \quad \text{where } \boldsymbol{\mu} = \begin{pmatrix} \mu_X \\ \mu_Y \end{pmatrix}, \quad \boldsymbol{\Sigma} = \begin{pmatrix} \sigma_X^2 & \rho\sigma_X\sigma_Y \\ \rho\sigma_X\sigma_Y & \sigma_Y^2 \end{pmatrix}$$

- PDF:

$$f(x,y) = \frac{1}{2\pi\sigma_X\sigma_Y\sqrt{1-\rho^2}} \exp\left(-\frac{1}{2\left(1-\rho^2\right)} \left[\frac{\left(x-\mu_X\right)^2}{\sigma_X^2} + \frac{\left(y-\mu_Y\right)^2}{\sigma_Y^2} - \frac{2\rho\left(x-\mu_X\right)\left(y-\mu_Y\right)}{\sigma_X\sigma_Y} \right] \right)$$

or

$$f_{\mathbf{X}}\left(x_1, \ldots, x_k\right) = \frac{\exp\left(-\frac{1}{2}(\mathbf{x}-\boldsymbol{\mu})^{\mathrm{T}}\boldsymbol{\Sigma}^{-1}(\mathbf{x}-\boldsymbol{\mu})\right)}{\sqrt{(2\pi)^k det(\boldsymbol{\Sigma})}}$$

Multivariate Normal Distribution (Contd)

Theorem

- If $(X, Y) \sim N_2(\mu, \Sigma)$, then X and Y are uncorrelated iff X and Y are independent
- Assume that $X \sim N_k(\mu, \Sigma)$ and $Y = A + BX$ $(A \in \mathbb{R}^l, B \in \mathbb{R}^{l \times k})$, then

$$Y \sim N_l(A + B\mu_X, B\Sigma B')$$

- Assume $X \sim N_k(\mu, \Sigma)$, let $Z = \Sigma^{-\frac{1}{2}}(X - \mu)$, then $Z \sim N_k(0, I_k)$, i.e. for each $i = 1, \ldots, k$, $Z_i \sim N(0, 1)$ and for $i \neq j$, $Cov(Z_i, Z_j) = 0$

How can we get $\Sigma^{-\frac{1}{2}}$? Because Σ is real-valued symmetric matrix, then $\exists$ orthogonal matrix[3] Q and diagonal matrix Λ whose entries are the eigenvalues of Σ s.t.

$$\Sigma = Q\Lambda Q'$$

[3] Here orthogonal means $Q'Q = I$. Moreover, columns of Q are the eigenvectors of Σ.

Multivariate Normal Distribution* (Counter Example)

- Two normal distributions (not jointly) with zero correlation do NOT imply independence
- Suppose that $X \sim N(0,1)$, $W = \pm 1$ w.p. $\frac{1}{2}$ and $Y = WX$. Also assume W and X are independent, then

$$Cov(X,Y) = \mathbb{E}[XY] - \mathbb{E}[X]\mathbb{E}[Y] = \mathbb{E}\left[X^2 W\right] - 0 = \mathbb{E}\left[X^2\right]\mathbb{E}[W] = 0$$

$$\begin{aligned}
P(Y \le x) &= P(\{WX \le x\} \cap (\{W = 1\} \cup \{W = -1\})) \\
&= P\left((\{WX \le x\} \cap \{W = 1\}) \cup (\{WX \le x\} \cap \{W = -1\})\right) \\
&= P\left(\{WX \le x\} \cap \{W = 1\}\right) + P\left(\{WX \le x\} \cap \{W = -1\}\right) \\
&= P\left(WX \le x | W = 1\right) P(W = 1) \\
&\quad + P\left(WX \le x | W = 1\right) P(W = -1) \\
&= P(X \le x)P(W = 1) + P(-X \le x)P(W = -1) \\
&= \frac{1}{2}\Phi(x) + \frac{1}{2}\Phi(x) = \Phi(x) \text{ which is the CDF of N(0,1)}
\end{aligned}$$

Chi-Squared Distribution

Recall that, if $X \sim N_k(\mu, \Sigma)$ where Σ is p.d. covariance matrix, then $Z = \Sigma^{-\frac{1}{2}}(X - \mu) \sim N_k(0, I_k)$ and

$$Y \equiv Z'Z = \sum_{i=1}^{k} Z_i^2 \sim \chi^2(k)$$

By independence of each Z_i and linearity, it can be shown that

- $\mathbb{E}[Y] = k$
- $Var(Y) = 2k$
- If $Y_1 = \sum_{i=1}^{m} Z_i^2 \sim \chi^2(m)$ and $Y_2 = \sum_{i=1}^{n} Z_i^2 \sim \chi^2(n)$ where $n > m$, then

$$Y_2 - Y_1 = \sum_{i=m+1}^{n} Z_i^2 \sim \chi^2(n-m)$$

t-Distribution

If $Z \sim N(0,1)$ and $Q \sim \chi^2(r)$ are independent, then we define t-distribution with r degrees of freedom as follow,

$$t \equiv \frac{Z}{\sqrt{Q/r}} \sim t_r$$

Some properties:

- Mean $\mathbb{E}[t] = 0$ and median $m = 0$
- $Var(t) = \begin{cases} \frac{r}{r-2} & \text{if } v > 2 \\ \infty & \text{if } 1 < v \leq 2 \\ undefined & \text{otherwise} \end{cases}$
- $t_\infty \sim N(0,1)$ (By WLLN and Slutsky's Theorem)

t-Distribution (Figure)

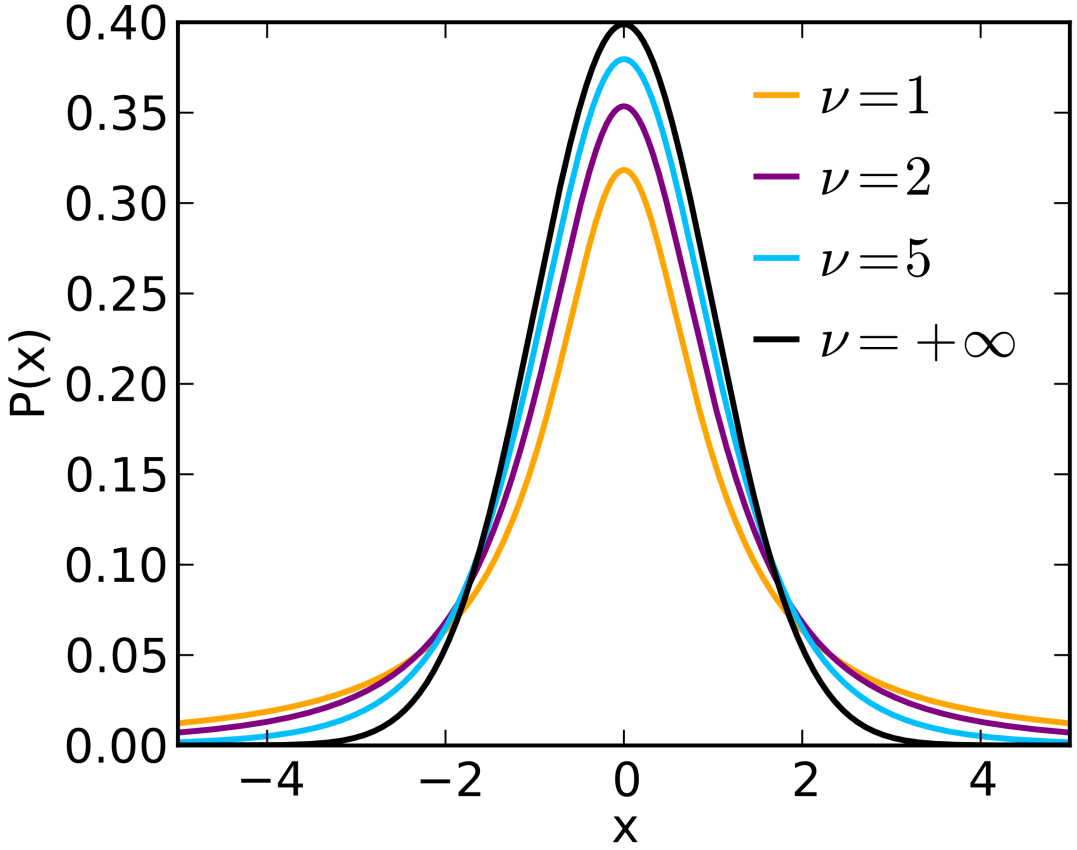

Figure: PDF of student's t-distribution t_ν (from Wikipedia)

F-Distribution

Let $Q_m \sim \chi^2(m)$ and $Q_r \sim \chi^2(r)$ be independent, then we can define F-distribution with degrees of freedom m and r, i.e. $F(m,r)$, by using Chi-squared distribution,

$$F \equiv \frac{Q_m/m}{Q_r/r} \sim F(m,r)$$

Interesting properties,

- If $F \sim F(m,r)$, then $F^{-1} \sim F(r,m)$
- If $t \sim t_r$, then $t^2 \sim F(1,r)$ and $t^{-2} \sim F(r,1)$
- If $F \sim F(m,r)$, then $mF \xrightarrow{d} \chi^2(m)$ when $r \longrightarrow \infty$

F-Distribution (Figure)

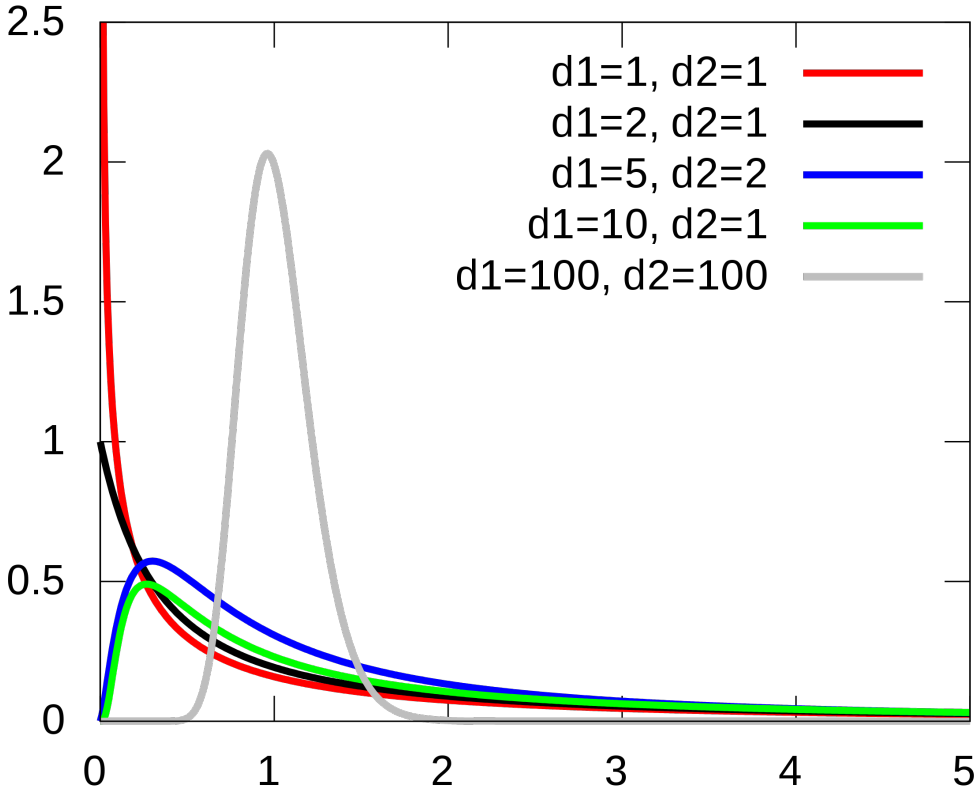

Figure: PDF of F-distribution $F(d_1, d_2)$ (from Wikipedia)